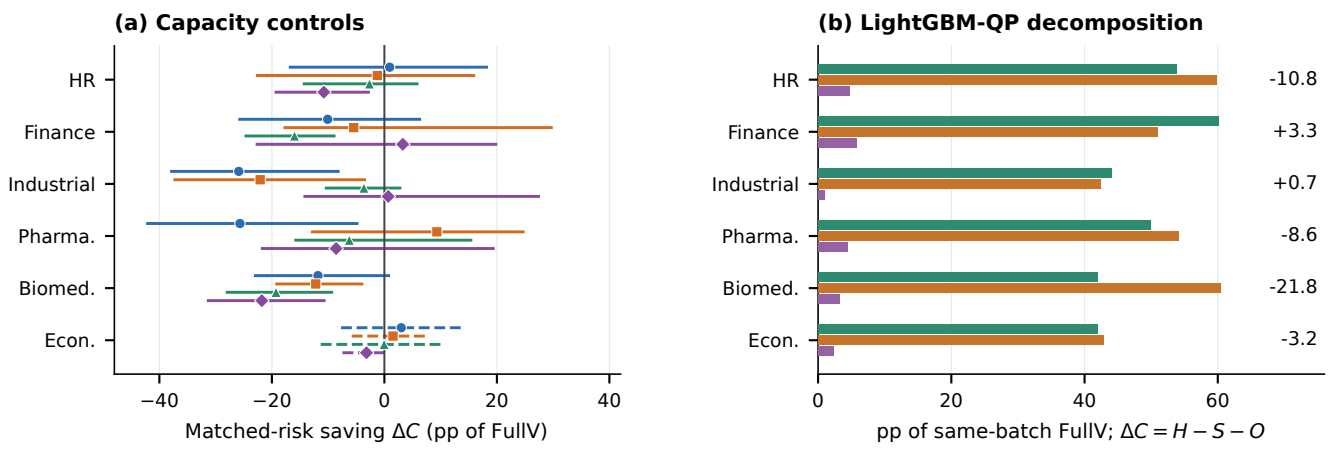
(a) Capacity controls
HR
Finance
Industrial
Pharma.
Biomed.
Econ.
−40
−20
0
20
40
Matched-risk saving ΔC (pp of FullV)
(b) LightGBM-QP decomposition
HR
Finance
Industrial
Pharma.
Biomed.
Econ.
-10.8
+3.3
+0.7
-8.6
-21.8
-3.2
0
20
40
60
pp of same-batch FullV; ΔC = H − S − O
Ridge Q
Ridge QP
LightGBM Q
LightGBM QP
H: headroom
S: selection loss
O: path difference